- This is a **Mirror Exam**: the same seven question types as the posted Practice Exam 2, with new numbers. Time yourself: **50 minutes**.
- The exam covers Chapters 4–6.
- The exam is **closed-book and closed-notes**.
- You may use a **calculator** and **one double-page cheat sheet**.
- **You must bring your physical UCF ID card. Your exam will not be graded if you fail to present your physical UCF ID card.**
- You are **not allowed** to use laptops, cellphones, tablets, or any other electronic devices.
- All cellphones must be **turned off** before the exam begins.
- Show all necessary steps for full credit.
- **Write all work clearly.** Answers without supporting steps may not receive credit and Round all probabilities to **four decimal places**.
- **Write answers in the spaces provided.** If you need additional space, clearly label the continuation.
- All needed z-scores and standard normal areas are given as **hints** on the exam paper.
- **Total points: 50.**

Question 0: Academic Integrity Acknowledgment (Required)

As your first task, please **handwrite the acknowledgment statement below** on your answer sheet. Then write your **full name**, **signature**, and **UCF ID number (in digits)** below it.

Acknowledgment Statement:

> "I acknowledge that I have read and understood all exam instructions and that I have fully complied, and will continue to comply, with the university's policies on Academic Integrity in completing and submitting this exam."

Name (print clearly): _____

Signature: _____

UCF ID Number: _____

Question 1 (6 points)

Commuting and Housing

The table below summarizes how 110 students at a university commute and where they live.

Commute	Lives on campus	Lives off campus
Drive	8	34
Bus	10	16
Walk	22	4
Bike	9	7

Total on campus = 49, Total off campus = 61, Overall Total = 110

Answer each question. Show your work when appropriate.

(a) **(1 points)** What is the probability that a randomly selected student lives off campus?

(b) **(1 points)** What is the probability that a randomly selected student walks to campus?

(c) **(1 point)** What is the probability that a randomly selected student drives OR lives on campus?

(d) **(1 points)** What is the probability that a randomly selected student bikes AND lives off campus?

(e) **(2 points)** What is the probability that a randomly selected student takes the bus given that the student lives off campus?

Question 2 (2 points): Discrete Probability Distribution

Determine whether the following distribution represents a valid **discrete probability distribution**. If not, explain why.

x	$P(x)$
1	0.15
2	0.25
3	0.30
4	0.20
5	0.12

Question 3 (8 points): Cars per Household

A city planner records the number of cars, X, owned by each household in a random sample of 80 households in one neighborhood. The following data represent the number of cars per household. *Note: no household in the sample owned more than 3 cars.*

x	Frequency
0	12
1	28
2	32
3	8

(a) **(2 pt)** Construct a probability distribution for the random variable X, the number of cars per household.

(b) **(2 pt)** Graph the discrete probability distribution.

(c) **(2 pt)** Compute and interpret the mean of X.

(d) **(2 pt)** Compute the standard deviation of X.

Question 4 (8 points): Reading Habits

According to a national survey, 35% of U.S. adults read at least one book last month. In a random sample of 8 U.S. adults, find the probability that:

(a) **(2 pt)** Exactly 2 read at least one book last month.

(b) **(2 pt)** At most 2 read at least one book last month.

(c) **(2 pt)** At least 1 read at least one book last month.

(d) **(2 pt)** For a random sample of 300 U.S. adults, find the expected number and standard deviation of those who read at least one book last month.

Question 5 (5 points): Shuttle Wait Time

Suppose the waiting time X (in minutes) for a campus shuttle is uniformly distributed between 0 and 25 minutes.

(a) **(1 pt)** Sketch or describe the graph of the uniform probability density function (pdf).

(b) **(2 pt)** Find the probability that the wait is between 5 and 15 minutes, i.e., $P(5 \le X \le 15)$.

(c) **(2 pt)** Find the probability that the wait is more than 18 minutes, i.e., $P(X > 18)$.

Question 6 (6 points): Normal Distribution and z-Scores

Instructions: Use the hints below (or the standard normal table) to find each probability. Show work and round final answers to four decimal places.

Hints: $\mathbf{P(Z < 0.85) = 0.8023}$, $\mathbf{P(Z < -1.30) = 0.0968}$, $\mathbf{P(Z < -0.25) = 0.4013}$, $\mathbf{P(Z < 1.25) = 0.8944}$.

(a) Find the area to the right of $z = 0.85$.

(b) Find the area between $z = -1.30$ and $z = -0.25$.

(c) The random variable X is normally distributed with $\mu = 250$ and $\sigma = 40$. Find $P(X > 300)$.

Question 7 (8 points): Heights of Adult Men

The heights of adult men in the U.S. are approximately normally distributed with mean $\mu = 69$ inches and standard deviation $\sigma = 2.8$ inches.

Hints: $\mathbf{P(Z < 1.79) = 0.9633}$, $\mathbf{P(Z < -1.43) = 0.0764}$, $\mathbf{P(Z < -1.07) = 0.1423}$, $\mathbf{P(Z < 1.07) = 0.8577}$; the 10th percentile corresponds to $\mathbf{z = -1.28}$.

(a) **(2 pt)** What is the probability that a randomly selected man is taller than 74 inches?

(b) **(2 pt)** What is the probability that a randomly selected man is shorter than 65 inches?

(c) **(2 pt)** What proportion of men are between 66 and 72 inches tall?

(d) **(2 pt)** What height places a man at the 10th percentile?